Autonomous Drone System for Ocean Monitoring Device Deployment and Retrieval

Version
Version 1.0

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Submission Date
2026.05.12

1. Product Overview

1.1 System Description

Seahawk is a hybrid drone-assisted system designed to deploy and retrieve environmental monitoring devices in near-shore environments. The system combines an aerial drone platform, computer vision tracking, a precision winch mechanism, and a servo-driven gripper to perform safe and repeatable payload operations without requiring human operators to enter hazardous coastal areas.

The system enables researchers to place and recover sampling devices from water surfaces while the drone remains in hover, eliminating the need for landing on unstable terrain or water.

1.2 Problem Addressed

Coastal environmental monitoring often requires deploying instruments in locations that are difficult or dangerous for humans to access. Traditional deployment methods rely on boats or manual placement, which introduces risks from tides, waves, slippery terrain, and changing weather conditions.

Seahawk reduces these risks by allowing remote deployment and retrieval using an autonomous drone platform equipped with a controlled lowering system and a precision gripping mechanism.

1.3 Target Users

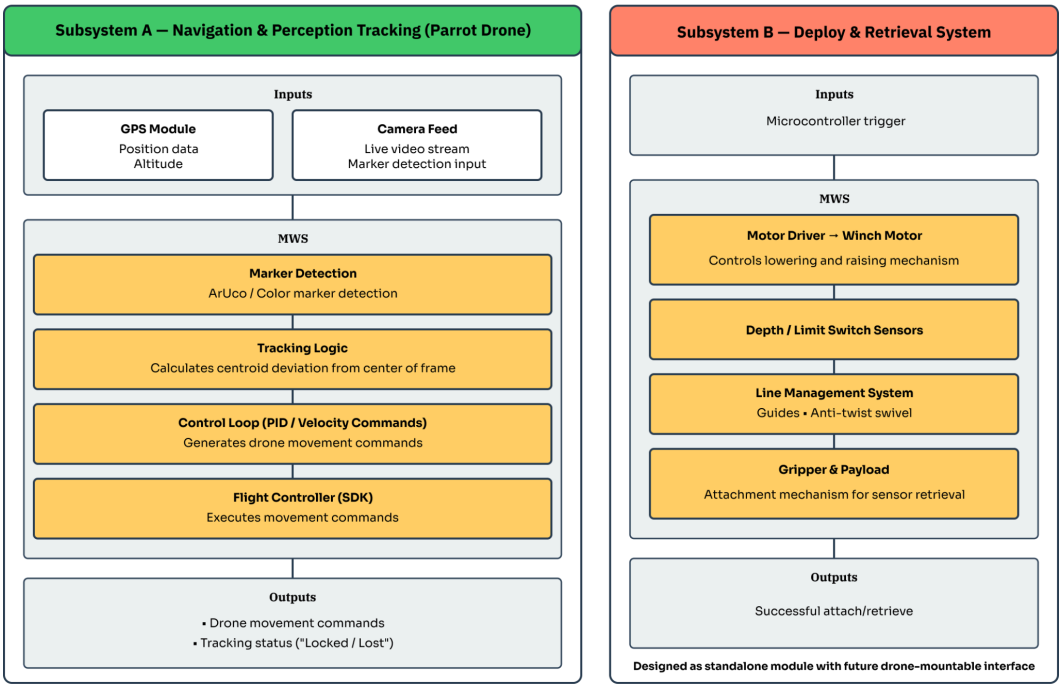
- Marine scientists
- Environmental researchers
- Coastal monitoring teams
- Field engineers

1.4 Primary Features

- Autonomous GPS navigation to deployment locations
- Visual target detection and tracking using computer vision
- Winch-based payload lowering system
- Rotating pinch-wheel gripper for deployment and retrieval
- Web-based control interface for system monitoring and control
- LoRa-based remote device location relay
- Closed-loop visual alignment for accurate positioning

2. System Overview

SeaHawks Subsystem Map



3. System Requirements

Software Requirements

Component	Requirement
Operating System	Ubuntu 20.04+ or Debian-based Linux
Python	3.8 or higher
Drone SDK	Parrot Olympe SDK
Computer Vision	OpenCV 4.5+
AI Detection	YOLOv8
GUI Framework	Flask + Vue.js
ESP Firmware	Arduino framework for ESP32

Internet is not required during operation but is required during initial installation for dependency downloads.

Hardware Requirements

Component	Description
Drone Platform	Parrot ANAFI Ai
Winch Controller	ESP32 microcontroller
Gripper Controller	ESP32 microcontroller
Camera	Drone integrated camera
Sensors	Hall sensor and limit switch
Communication	WiFi + LoRa (optional)
Power Supply	Drone battery + external 5V supply for ESP modules

Environmental Conditions

Operating conditions:

- Temperature: 0°C – 40°C
- Wind Conditions: Recommended below 10 m/s
- Weather: Operation should not occur in rain or heavy fog.

Package Contents

The Seahawk system includes:

- Drone platform
- Winch assembly
- Gripper module with housing
- ESP32 winch controller
- ESP32 gripper controller
- Hall sensor cable
- Limit switch cable
- Mounting hardware
- USB cable for ESP programming

4. Installation and Setup

4.1 Software Installation

Clone the repository:

```
git clone https://github.com/GIX-Luyao/final-codebase-seahawks.git
```

Install dependencies:

```
pip install -r requirements.txt
pip install -r gui/requirements_gui.txt
```

Install Parrot Olympe SDK:

```
pip install parrot-olympe
```

4.2 ESP32 Firmware Configuration

Both the winch and gripper controllers run ESP32 firmware that connects to the drone WiFi network.

Before uploading the firmware, configure the WiFi credentials in the ESP source file.

Example:

```
const char* ssid = "ANAFI Ai XXXXX";
const char* password = "XXXXXXXX";
```

Upload the firmware using the Arduino IDE or PlatformIO.

4.3 Hardware Assembly

Winch Assembly

The winch module contains:

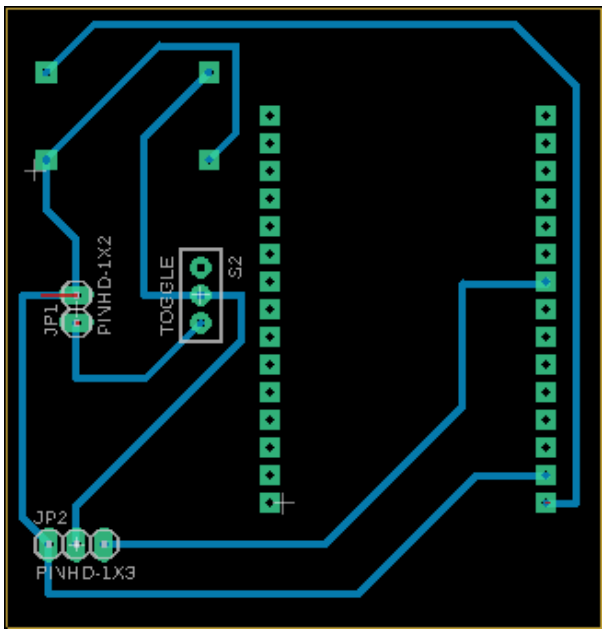
- continuous rotation servo
- cable spool
- Hall effect sensor for cable length measurement
- limit switch for over-pull protection

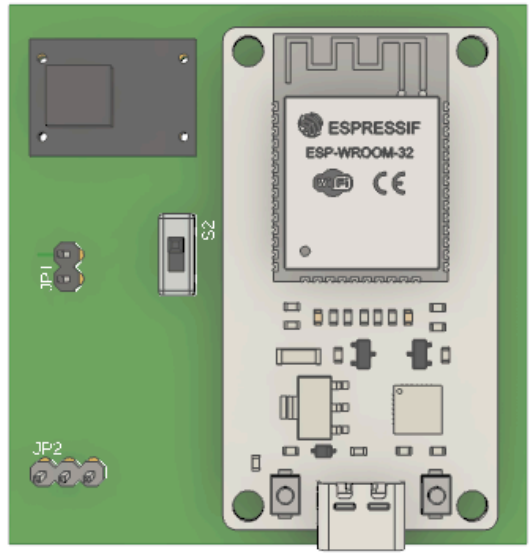
The design follows principles similar to 3D-printed winch systems used for drone payload operations

Gripper Assembly

The gripper module uses a rotating pinch-wheel design to perform three operations:

1. Hold payload during flight
2. Release payload during deployment
3. Re-grip payload during retrieval





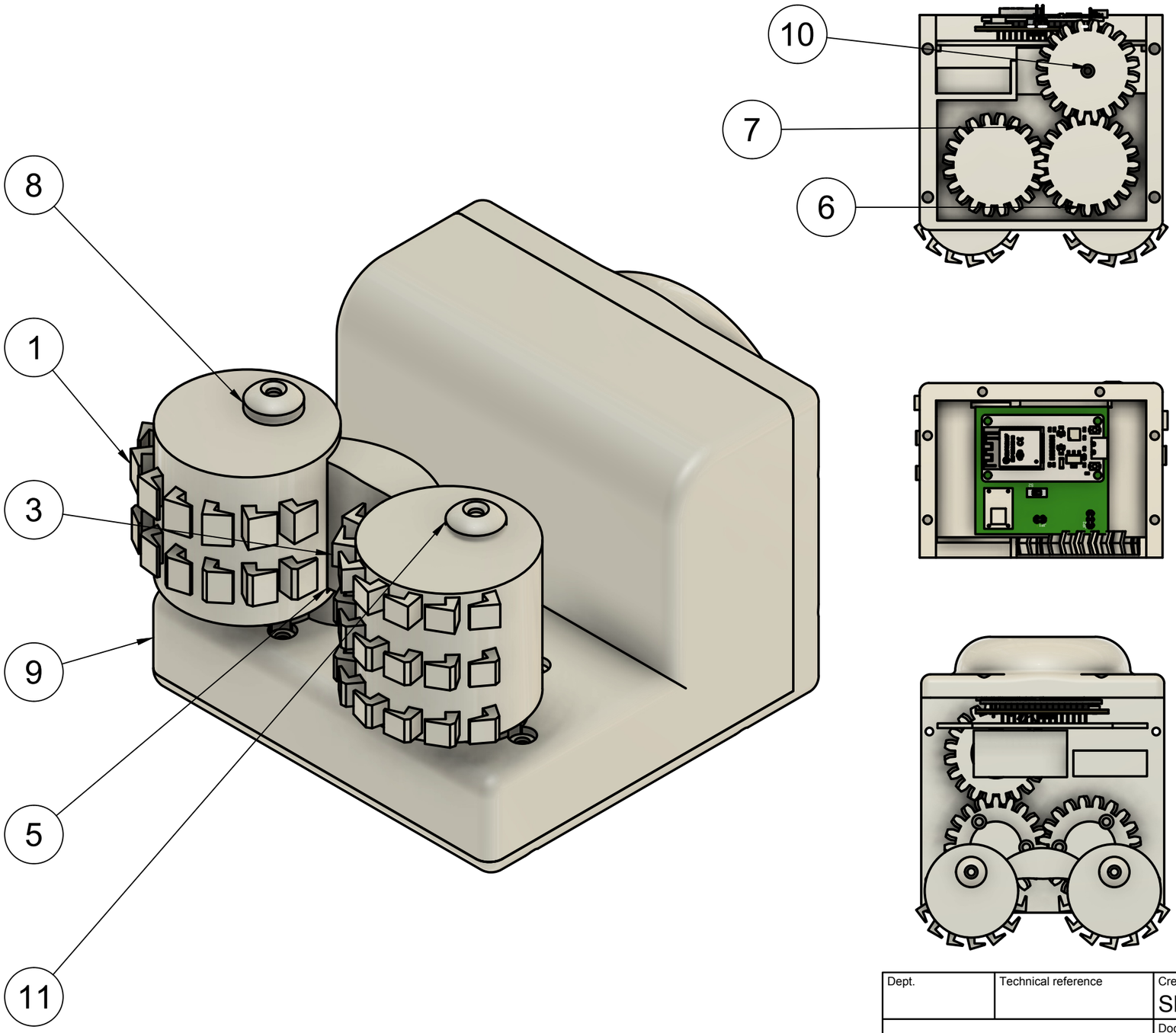
5. Operating Instructions

5.1 Starting the System

1. Power on the drone.
2. Connect the control laptop to the drone WiFi network.
3. Power on ESP32 winch and gripper modules.
4. Launch the control GUI:
   ```
   cd gui
   python3 drone_control_gui.py
   ```
5. Open the interface at: http://localhost:5000

5.2 Deployment Operation

1. Take off the drone.
2. Navigate to the deployment location using GPS waypoint control.
3. Activate visual tracking mode.
4. Align the drone above the deployment target.
5. Activate winch lowering.



Parts List			
Item	Qty	Part Number	Material
1	1	Gripper	Steel
2	1	15' Connecter	Steel
3	1	Gripper(Mirror)	Steel
4	1	15' Connector(Mirror)	Steel
5	1	Mock Device Clip v2	Steel
6	1	40' DHex Gear	Steel
7	1	40' DHex Gear R	Steel
8	1	Cap	Steel
9	1	Gripper Motor Housing	----
10	1	40' DHex Gear Motor	Steel
11	1	Cap (1)	Steel

Dept.	Technical reference	Created by Shih-Yu Wu 2026/3/13	Approved by
		Document type	Document status
		Title Gripper v2	DWG No.
			Rev. / Date of issue / Sheet 1/9

6. Release payload using gripper command.

5.3 Retrieval Operation

1. Navigate drone to the device location.
2. Activate visual detection to center the drone above the target.
3. Lower the winch cable.
4. Close the gripper to capture the device.
5. Pull the cable to retrieve payload.

5.4 Shutdown Procedure

1. Land the drone.
2. Disconnect GUI session.
3. Power off ESP modules.
4. Turn off the drone.

6. Technical Specifications

Drone Platform

Specification	Value
Drone	Parrot ANAFI Ai
Communication	WiFi
Camera	Integrated 48MP camera

Winch System

Specification	Value
Cable Spool Diameter	~15.7 mm
Cable Length Measurement	Hall sensor pulse counting
Motor Type	Continuous rotation servo
Safety Feature	Limit switch over-pull protection

Gripper System

Specification	Value
Actuation	Servo motor
Grip Type	Rotating pinch wheel
Housing	3D printed enclosure

Communication Protocols

Link	Protocol
Laptop → Drone	WiFi
Laptop → Winch ESP32	HTTP
Laptop → Gripper ESP32	WebSocket
LoRa Device → Base	LoRa

7. Safety Considerations

Electrical Safety

- Ensure all wiring connections are insulated and secure before powering the system.

Mechanical Safety

- The winch cable may move suddenly during operation.
- Operators must keep hands clear of the spool and gripper mechanism while the system is active.

Flight Safety

- Drone operations must follow FAA regulations.
- The drone must not operate above people or in restricted airspace.

Environmental Risks

- Strong winds can destabilize suspended payloads.
- Operations should be suspended in wind conditions exceeding safe limits.

8. Troubleshooting

Issue	Possible Cause	Solution
Drone cannot connect	Not on drone WiFi	Connect to ANAFI network
Winch not responding	ESP32 offline	Verify WiFi connection
No target detection	Lighting mismatch	Recalibrate HSV thresholds
Navigation failure	Low GPS signal	Move to open outdoor area

9. Limitations

The system currently has several known limitations:

- Visual detection performance may vary depending on lighting conditions and reflections on water surfaces.
- The drone payload capacity limits the size and weight of deployable monitoring devices.
- The system currently supports tracking only one target at a time.
- Autonomous landing on water surfaces is not supported.
- Strong wind conditions may reduce winch stability and payload alignment.